

SIMRA ALI

Westford Academy | Class of 2027 | Westford, Massachusetts | [linkedin.com/in/sim-ali](https://www.linkedin.com/in/sim-ali)

EDUCATION

Westford Academy

Class of 2027

Intended field: Computer Engineering / Electrical & Computer Engineering; interests in robotics, autonomous systems, embedded systems, and human-robot interaction.

- GPA: 4.19 | SAT: 1560 (790 Math, 770 EBRW).
- Selected coursework: AP Calculus BC, AP Physics, AP Chemistry, AP Computer Science A, AP Biology, AP U.S. History, AP Psychology, AP Microeconomics.

ROBOTICS LEADERSHIP & ENGINEERING

FIRST Robotics Competition — Stormgears 5422 (45 members)

2023–Present

Co-Captain • Drive Team Coach • Lead CAD/Mechanical Designer • Former Electrical Lead

- Progressed from electrical to multidisciplinary technical leadership; designed competition-robot mechanisms in SolidWorks, including chassis, intake, hopper, shooter, and serviceable electrical systems.
- Self-taught SolidWorks after the team’s experienced designer graduated; helped lead a custom-robot effort that earned the FIRST Quality Award for design intent, robustness, and fabrication quality.
- Organize technical trainings and mentor new members; trained a younger student who later succeeded me as Electrical Lead. Served as Competition Technician and Coach under match pressure.

FIRST Tech Challenge — Ghost Robotics 3565 (10 members)

2023–Present

Vision Software • CAD/Mechanical Design • Media • Drive Team

- Contribute across Onshape CAD, computer vision, strategy, outreach, and team needs; intentionally work across subteams rather than limiting contributions to one formal title.
- 2026 Massachusetts State Champion and Winning Alliance Captain; 2026 FIRST Championship Ross Division Winner. Team earned Control, Inspire, Innovate, and Connect Awards.

ADVANCED ENGINEERING & RESEARCH

MIT Beaver Works Summer Institute — Autonomous Air Vehicle Racing

Summer 2026

Four-week in-person program, MIT AeroAstro

- Designed, programmed, repaired, tested, and raced autonomous drones; studied computer vision, state estimation, control systems, navigation, and flight dynamics.
- Explored socially aware path planning for autonomous drones and connected with an MIT PhD researcher in the field; BWSI “You Belong in STEM” Essay Contest Honorable Mention.

MIT Beaver Works — Technical Coursework

2024–2026

- Completed Basics of ASICs, Microelectronics & Hardware Development, Python Core, and Git/GitHub.

Harvard Summer School

Summer 2025

Robotics, Autonomous Vehicles, Drones, and Artificial Intelligence — Grade: A

- Studied sensors, motion planning, SLAM, reinforcement learning, autonomous systems, and computer vision; co-led “Solve It, Defend It!” and developed the Flood Watch concept using drone/AV data.

BAE Systems Women in Technology

Jan–May 2026

- Completed hands-on labs, workshops, tours, and presentations spanning electrical, microwave/RF, software, mechanical, signal processing, optical, manufacturing, and failure-analysis engineering.

INNOVATION, OUTREACH & ACTIVITIES

Women in Computer Science — Westford Academy

Co-Founder / Co-Lead

- Teach Python, support projects and hackathon preparation, share STEM opportunities, and build a supportive community for students exploring computing.

Hackathons & Engineering Competitions

2025

- Microsoft NERD GirlHacks at Microsoft NERD — 1st Place: built “The Bat,” an ultrasonic assistive navigation device for visually impaired users.
- NetScout Tech for Good — 1st Place: developed HireAble, a platform connecting people with disabilities to inclusive employers.
- VTXCADs CADathon, Phillips Exeter Academy Robotics — 1st Place.

STEM Outreach & Mentorship

Ongoing

- Lead and support Girl Scout STEM Fest, FIRST LEGO League trainings/scrimmages, Apple Blossom Parade, school/community demonstrations, and robotics workshops; emphasize mentoring and retaining girls in technical roles.

Westford Academy Broadcasting Club

2025–Present

Manager; lead and coordinate club operations while contributing as an anchor and production crew member, with experience in teleprompter, sound board, video switching, scripts, and school-wide broadcasts.

SELECTED HONORS

FIRST Leadership Award (Dean’s List) Semi-Finalist (2026) • MIT BWSI “You Belong in STEM” Honorable Mention (2026) • Microsoft NERD GirlHacks 1st Place (2025) • NetScout Tech for Good 1st Place (2025) • VTXCADs CADathon 1st Place (2025) • Westford Academy Academic High Honors (2024–2026)

TECHNICAL SKILLS

CAD/Mechanical: SolidWorks, Onshape, 3D printing | **Hardware:** Arduino, wiring, soldering, crimping, electrical-board layout, sensors, troubleshooting | **Software:** Python, Java, Git/GitHub, HTML/CSS/JavaScript, WPILib basics | **Robotics/AI:** computer vision, SLAM, reinforcement learning, autonomous systems